J8: 1=3V3 VTref, 2=SWDIO, 3=GND
4=SWCLK, 5=NRST, 6=SWO
VTref: target voltage sense, no debugger power injection.

USB FS DEVICE / SELF-POWERED
USB_VBUS isolated from board +5V (no backfeed).
PA9: GPIO VBUS detect; enable USB only with VBUS.
VDDUSB = +3V3; enable PWR_CR2.USV in firmware.
Internal D+ pull-up / termination. 90-ohm differential PCB pair.

MCU: internal HSI/MSI; USB HSI48 + CRS/SOF.
PH0/PH1: no external HSE. PC14/PC15 are GPIO, no LSE.
BOOT0 = PH3 pin44 (PB8_BOOT); PB8 pin45 = DRV_AGC0.
VDDA/VREF+ share pin9; disable VREFBUF output.
C38: VBAT 100nF; C39: VDDA 10nF; C40: VDDUSB 1uF.
BOOT0 uses existing boot switch/pull-down (PB8_BOOT net name is legacy).
Place bypass capacitors close to their assigned MCU pins.
Use HSI/MSI internal clock for USB standalone startup.
NC GPIOs: PA1, PA7, PB2, PA15, PB4, PB5.

POWER INPUT 12–24V (J6) / reverse polarity + transient protection

VM sense: 12–24V input, 1/19 divider, powered-off isolation.
PA0 = ADC1_IN5; VM = ADC/4095 * VDDA * 19.
Wait >=10ms after power switching; ADC sample >=247.5 cycles.
ADDR2 (PC13): R58 pulls low, default address bit 0.

AS5600 external sensor / 3.3V I2C
STM32 I2C1: PB6=SCL, PB7=SDA (AF4)
J5: 1=3V3, 2=GND, 3=SCL, 4=SDA
Fit one set of pull-ups on the bus.

A=PA15/TIM2_CH1, B=PA1/TIM2_CH2 (AF1).
3.3V logic; GPIO/EXTI for home inputs.
Use MCU pull-ups for dry contact/open-drain sensors.

STM32L433 IC4 <–> TB67S249 IC2 – stm32l433_tb67s249_connection.md

AGC straps: R50=0; R52/R54/R56=100k; R51/R53/R55/R57=DNP.
CLIM1/BOOST/FLIM pull-ups must use DRV_VCC, never +3V3_D.

LO1/LO2/MO pull-ups: +3V3_D only (external digital 3.3V supply).
VM and +3V3_D supplied by power circuitry outside this specification.

OSCM: R23=5.1k, C20=270pF; ~67kHz.
Place close to pin35. DRV_VCC: internal 5V, setting circuits only.

VREF: R29=10k series, R30=9.1k to GND, C21=100nF.
3.3V DAC full scale –> VREF 1.57V –> IOUT 1.97A.
R30/C21 GND return near pins29/30, away from OUT/VM.

VM capacitors: C15 bulk 220uF/63V; C16 at VMA; C17 at VMB; C18 10uF/50V.
Keep VM–PGND loops short; paired pins joined at IC on PCB.

NT1: single PGND–GND copper connection directly beneath IC2 exposed pad.
EP/corner pads to GND, dense thermal vias; 4 layers / 2oz. Motor outputs: 5A routing.

Firmware: DAC1_OUT1 PA4 buffer enabled; reference VDDA/VREF+ (+3V3_A).
Buffered DAC may retain ~0.2V: ENABLE Low or disable DAC for zero current.
Startup: check VM –> set DAC –> set DMODE –> wait >=7.5us –> ENABLE High.
Reset: ENABLE/DMODE/AGC Low; evaluate initially with AGC0=AGC1=Low.

USB only: MCU on, motor VM off.
USB current before enumeration <=100mA total.
VM = ADC/4095 * VDDA * 19; wait >=10ms after power changes.
4.5A is driver absolute peak maximum; initial VREF limit ~1.97A/phase.
Route high–current PGND via NT1 only; keep BUCK_SW away from ADC.
AP2112 dissipation and motor/Q1 cooling require thermal validation.

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